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Computer Aided Verification

23rd International Conference, CAV 2011
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Proceedings



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Preface

The International Conference on Computer-Aided Verification (CAV) is dedicated to the advancement of the theory and practice of computer-aided formal analysis methods for hardware and software systems. Its scope ranges from theoretical results to concrete applications, with an emphasis on practical verification tools and the underlying algorithms and techniques. This volume contains the proceedings of the 23rd edition of this conference held in Snowbird, Utah, USA, during July 14–20, 2011. The conference included two workshop days, a tutorial day, and four days for the main program.

At CAV 2009, Bob Kurshan approached us with the idea of holding CAV 2011 in Salt Lake City. Encouraged by the enthusiastic support from late Amir Pnueli, we had little hesitation in agreeing to Bob’s proposal. While the initial proposal was to organize the conference on the campus of the University of Utah, we eventually decided to hold it at the Snowbird resort near Salt Lake City. Our decision was motivated by the dual desire to showcase the abundant natural beauty of Utah and to provide a collegial atmosphere similar to a Dagstuhl workshop.

We are happy to report that CAV is thriving, as evidenced by the large number of submissions. We received 161 submissions and selected 35 regular and 20 tool papers. We appreciate the diligence of our Program Committee and our external reviewers due to which all (except two) papers received at least four reviews. A big thank you to all our reviewers!

The conference was preceded by the eight affiliated workshops:

- The 4th International Workshop on Numerical Software Verification (NSV 2011), Thursday, 7/14
- 10th International Workshop on Parallel and Distributed Methods in Verifications (PDMC 2011), Thursday, 7/14
- The 4th International Workshop on Exploiting Concurrency Efficiently and Correctly (EC² 2011), 7/14-7/15
- Frontiers in Analog Circuit Synthesis and Verification (FAC 2011), 7/14-7/15
- International Workshop on Satisfiability Modulo Theories, including SMT-COMP (SMT 2011), 7/14-7/15
- 18th International SPIN Workshop on Model Checking of Software (SPIN 2011), 7/14-7/15
- Formal Methods for Robotics and Automation (FM-R 2011), 7/15
- Practical Synthesis for Concurrent Systems (PSY 2011), 7/15

In addition to the presentations for the accepted papers, the conference also featured four invited talks and four invited tutorials.

- Invited talks:
 - *Andy Chou* (Coverity Inc.): “Static Analysis Tools in Industry: Notes from the Front Line”

- *Vigyan Singhal* and Prashant Aggarwal (Oski Technology): “Using Coverage to Deploy Formal Verification in a Simulation World”
- *Vikram Adve* (University of Illinois at Urbana-Champaign): “Parallel Programming Should Be and Can Be Deterministic-by-default”
- *Rolf Ernst* (TU Braunschweig): “Formal Performance Analysis in Automotive Systems Design: A Rocky Ride to New Grounds”
- Invited tutorials:
 - *Shuvendu Lahiri* (Microsoft Research): “SMT-Based Modular Analysis of Sequential Systems Code”
 - *Vijay Ganesh* (Massachusetts Institute of Technology): “HAMPI: A String Solver for Testing, Analysis and Vulnerability Detection”
 - *Ranjit Jhala* (University of California at San Diego): “Using Types for Software Verification”
 - *Andre Platzer* (Carnegie Mellon University): “Logic and Compositional Verification of Hybrid Systems”

A big thank you to all our invited speakers!

We thank the members of the CAV Steering Committee —Michael Gordon, Orna Grumberg, Bob Kurshan, and Ken McMillan— for their timely advice on various organizational matters. Neha Rungta, our Workshop Chair, smoothly handled the organization of the workshops. Eric Mercer, our Local Arrangements Chair, set up the registration portal at Brigham Young University. Sandip Ray, our Publicity Chair, helped publicize CAV 2011. We thank Aarti Gupta, past CAV Chair, for her help and advice in running the conference and maintaining its budget.

We thank Geof Sawaya for maintaining the CAV 2011 website. We are grateful to Wendy Adamson for arranging the beautiful Cliff Lodge facility at an affordable price and really making the budget work in our favor. We thank Alfred Hofmann of Springer for publishing the paper and USB proceedings for CAV 2011. We thank Andrei Voronkov and his team for offering us EasyChair which has proven invaluable at every juncture in conducting the work of CAV. We thank the office staff of the School of Computing, University of Utah, especially Karen Feinauer and Chris Coleman, for allowing us to use the school resources for managing CAV activities.

We are especially grateful to our corporate sponsors —Microsoft Research, Coverity, Google, NEC Research, Jasper, IBM, Intel, Fujitsu, and Nvidia— for their donations. We are also grateful to Judith Bishop and Wolfram Schulte of Microsoft Research for their substantial financial backing of CAV. We also thank Lenore Zuck, Nina Amla, and Sol Greenspan who helped with obtaining an NSF travel award.

CAV 2012 will be held in Berkeley, California.

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